Hawai‘i – Tsunami ESF-6 playbook

Data as of 2026-10-27

County: Hawai‘i · **Hazard:** Tsunami

Planning aid. Supports, does not replace, county emergency plans. Not an official shelter list.

1. Summary

Calculated by open code from public data · see Sources (section 8)

People in the risk area **12,462**	May need a public shelter (likely) **623**	May need a public shelter (low – high) **249 – 1,371**
Speak English less than well (C) **462**	Use power-dependent medical equipment (estimate) (M) **86**	People with a disability (I) **1,839**
Age 65 and older (I) **2,675**	People in households without a car (T) **864**	Known 2026 shelter sites usable for this hazard **10**
People in the extreme tsunami zone (worst case) **14,008**		

Source: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER, FEMA National Risk Index v1.20; shelter rates in section 8.

Share of people in the risk area who may need a public shelter: low 2%, likely 5%, high 11.0%. One person can be counted in more than one need. Page 2 explains how to read these numbers.

2. Scope: county total vs. one community

The county total is an upper bound. It counts the people living inside the tsunami evacuation zone as hit at the same time. Real events usually hit fewer communities.

Largest single community (the Census areas with the most people who may need a shelter)

Community (Census area)	People in the risk area	May need a public shelter (low – likely – high)	Speak English less than well	Use power-dependent medical equipment (estimate)	People with a disability	Age 65 and older	People in households without a car
Kailua Census area 216.08	1,971	39 – **99** – 217	150	16	264	487	27
Hawaiian Paradise Park Census area 210.14	1,391	28 – **70** – 153	44	8	143	257	0
Hilo Census area 205	1,344	27 – **67** – 148	36	8	202	236	291

Source: people and needs – CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER and FEMA National Risk Index v1.20; community names – 2020 Census blocks and places, spellings from USGS GNIS.

To plan for one community, use the first row.

How to read this playbook

Example row

Community (Census area)	People in the risk area	May need a public shelter (low – likely – high)
Kailua Census area 216.08	1,971	39 – **99** – 217

Read it as: about 1,971 people live in the risk area of Kailua. If this hazard hits, between 39 and 217 of them may need a public shelter; 99 is the likely number.

Census area

A neighborhood-sized area the U.S. Census uses, usually 1,200–8,000 people. The number (for example "Census area 308") is its Census ID. Each area is named after the Census place where most of its people live. "near" means no one in the area lives inside a named place; "rural near" means most people live outside named places.

Low – likely – high

The share of people in the risk area who may go to a public shelter: low 2%, likely 5%, high 11.0%. Where the rates come from: Team assumptions: likely 5%; low 2% (the hurricane planning factor); high 11% (the Lahaina 2023 rate).

Who has access and functional needs (CMIST)

- **C – Communication:** people who speak English less than well; they may need interpreters and translated messages.
- **M – Medical:** people who use power-dependent medical equipment (an estimate from HHS emPOWER).
- **I – Independence:** people with a disability and people age 65 and older.
- **S – Safety and support:** children and people living in group quarters (in the needs table).
- **T – Transportation:** people in households without a car.

County total vs. one community

The county total counts the people living inside the tsunami evacuation zone as hit at the same time, so it is an upper bound. Use the "Largest single community" table to plan for an event that hits one community.

Map

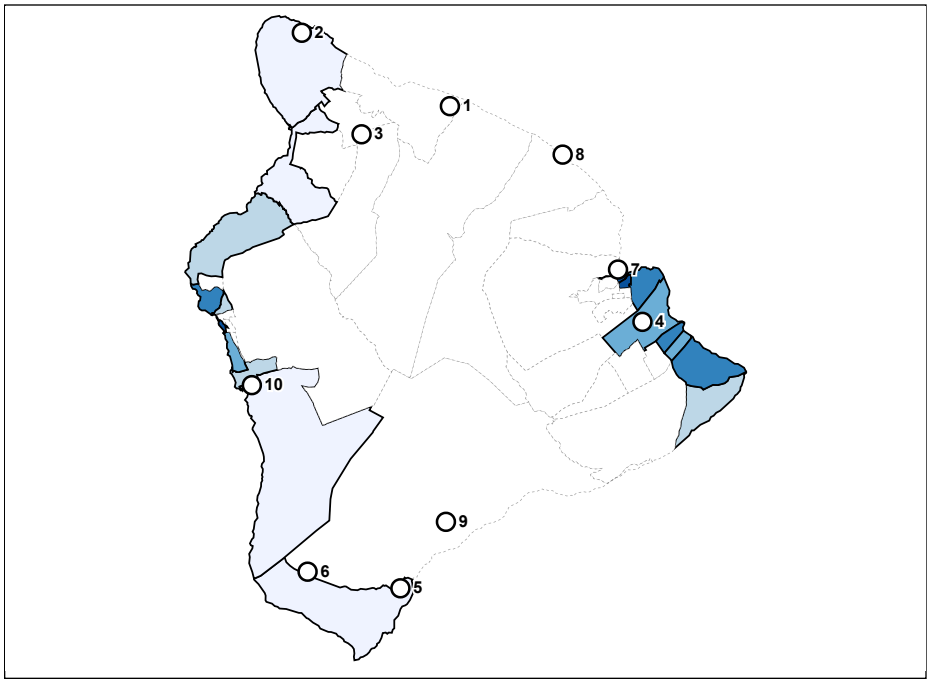
Darker blue means more people with needs, from "Lowest need" to "Highest need". A solid outline marks a Census area in the risk area; a dashed outline is outside it. Symbols: ○ usable site, △ check flood risk, ◇ flood risk not mapped, ⊠ not for this hazard. Numbers on the map match the site key.

Why population numbers can differ

People in the risk area come from ACS 2018–2022 estimates (through the CDC Social Vulnerability Index 2022). Community names use 2020 Census counts. County totals in the plan use Census estimates for 2025. Each table names its source underneath.

Every number is a planning estimate. In an event, follow county announcements.

3. Map: people with needs and known sites



Site key

1 = Honokaʻa Gym (usable)
2 = Ikuo Hisaoka Gym (usable)
3 = Kahilu Town Hall (usable)
4 = Keaʻau Armory (usable)
5 = Nāʻālehu Elementary School Gym (usable)
6 = Ocean View Community Center (usable)
7 = Puʻuʻeo Community Center (usable)
8 = Pāpaʻaloa Park Annex (usable)
9 = Robert N. Herkes Gym (usable)
10 = Yano Hall (usable)

- Lowest need (0–37 people)
- Lower (37–103 people)
- Middle (103–152 people)
- Higher (152–315 people)
- Highest need (315–537 people)
- Solid outline: Census area in the risk area
- Dashed outline: outside the risk area
- Usable site
- Check flood risk
- Flood risk not mapped (FEMA zone D)
- Not for this hazard

Color = people in the risk area who speak English less than well, use power-dependent medical equipment, have a disability, or live without a car (a person can be counted more than once). No street map: Census area outlines only.

Source: people and needs – CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER and FEMA National Risk Index v1.20; community names – 2020 Census blocks and places, spellings from USGS GNIS.

4. Known sites (opened in 2026 events)

Compiled from official county releases · verified by team

#	Status for this hazard	Site	Address	Why	Times opened in 2026
1	✓ **Usable**	Honokaʻa Gym	45-541 Lehua St., Honokaʻa	Outside the tsunami evacuation zone	1
2	✓ **Usable**	Ikuo Hisaoka Gym	54-382 Kamehameha Park Road, Kapaʻau	Outside the tsunami evacuation zone	1
3	✓ **Usable**	Kahilu Town Hall	67-1182 Lindsey Road, Waimea	Outside the tsunami evacuation zone	1
4	✓ **Usable**	Keaʻau Armory	16-512 Volcano Road, Keaʻau	Outside the tsunami evacuation zone	1
5	✓ **Usable**	Nāʻālehu Elementary School Gym	95-5545 Mamalahoa Highway, Nāʻālehu	Outside the tsunami evacuation zone	1
6	✓ **Usable**	Ocean View Community Center	92-8924 Leilani Circle, Ocean View	Outside the tsunami evacuation zone	1
7	✓ **Usable**	Puʻuʻeo Community Center	145 Wainaku St., Hilo	Outside the tsunami evacuation zone	1
8	✓ **Usable**	Pāpaʻaloa Park Annex	35-1994 Government Main Road, Pāpaʻaloa	Outside the tsunami evacuation zone	1
9	✓ **Usable**	Robert N. Herkes Gym	96-1219 Kamani St., Pāhala	Outside the tsunami evacuation zone	1
10	✓ **Usable**	Yano Hall	82-6156 Māmalahoa Highway, Captain Cook	Outside the tsunami evacuation zone	1

Source: county announcements of sites opened in 2026; FEMA flood zones (Hawaiʻi Statewide GIS, State DFIRM); tsunami evacuation zones (Hawaiʻi Statewide GIS).

Candidate public schools in this county usable for this hazard: **49**. Sites are places a county has opened before; not every site opens in every event. Always follow current county announcements.

5. Needs and proposed partners

Likely estimate. Roles are **proposed – to be confirmed with partner**. The pledged column stays blank until a partner confirms it.

Need	How many	Unit	Proposed lead	Proposed support	Pledged by partner	Still needed
Cots / sleeping space	624	cots	County Emergency Management	Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies	—	confirm with partner
Floor space (≤72 h)	12,462	sq ft	County Emergency Management	Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies	—	confirm with partner
Floor space (≤2 weeks)	24,924	sq ft	County Emergency Management	Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies	—	confirm with partner
Water	624	gal/day	American Red Cross	Hawai‘i Department of Human Services (DHS), County Emergency Management, Hawai‘i Department of Education (DOE) – schools, The Salvation Army, Hawai‘i VOAD (Voluntary Organizations Active in Disaster), Faith-based organizations, Business community	—	confirm with partner
Meals	1,247	meals/day	American Red Cross	Hawai‘i Department of Human Services (DHS), County Emergency Management, Hawai‘i Department of Education (DOE) – schools, The Salvation Army, Hawai‘i VOAD (Voluntary Organizations Active in Disaster), Faith-based organizations, Business community	—	confirm with partner
Toilets (≤72 h)	16	toilets	County Emergency Management	Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies	—	confirm with partner
Toilets (≤2 weeks)	32	toilets	County Emergency Management	Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies	—	confirm with partner
Shelter staff	46	staff per shift	County Emergency Management	Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies, Hawai‘i VOAD (Voluntary Organizations Active in Disaster), Office of Hawaiian Affairs (OHA), The Salvation Army, Philanthropy (foundations and donors), Business community	—	confirm with partner
Power outlets for power-dependent residents	5	outlets	Hawai‘i Department of Health (DOH)	County Emergency Management, Health care organizations (incl. managed health care), Utilities	—	confirm with partner
Accessible transport	864	people in no-vehicle households at risk	County Emergency Management	Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Transportation (DOT), Business community, Other state and county agencies	—	confirm with partner
Pet space – dog households	101	households with a dog	County Emergency Management	Hawai‘i VOAD (Voluntary Organizations Active in Disaster)	—	confirm with partner
Pet space – cat households	59	households with a cat	County Emergency Management	Hawai‘i VOAD (Voluntary Organizations Active in Disaster)	—	confirm with partner
Children's area	128	children in shelter	County Emergency Management	Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies, Hawai‘i Department of Health (DOH), Health care organizations (incl. managed health care), Philanthropy (foundations and donors)	—	confirm with partner
Interpreters – Ilocano	6	people with limited English in shelter	Hawai‘i Department of Human Services (DHS)	County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies	—	confirm with partner

Need	How many	Unit	Proposed lead	Proposed support	Pledged by partner	Still needed
Interpreters – Other English-based Creole languages	4	people with limited English in shelter	Hawai‘i Department of Human Services (DHS)	County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies	—	confirm with partner
Interpreters – Japanese	2	people with limited English in shelter	Hawai‘i Department of Human Services (DHS)	County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies	—	confirm with partner
Interpreters – Spanish	2	people with limited English in shelter	Hawai‘i Department of Human Services (DHS)	County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies	—	confirm with partner
Interpreters – Tagalog, Filipino	2	people with limited English in shelter	Hawai‘i Department of Human Services (DHS)	County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies	—	confirm with partner
Interpreters – Chinese	1	people with limited English in shelter	Hawai‘i Department of Human Services (DHS)	County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies	—	confirm with partner
Interpreters – Hawaiian	1	people with limited English in shelter	Hawai‘i Department of Human Services (DHS)	County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies	—	confirm with partner
Interpreters – Marshallese	1	people with limited English in shelter	Hawai‘i Department of Human Services (DHS)	County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies	—	confirm with partner

Source: likely estimate × planning factors in section 8 (Florida Division of Emergency Management 2010; AVMA 2022 for pets).

6. Decision points

Proposed – supports, does not replace, county plans.

When	Trigger	Action	Owner (proposed)	Support
BEFORE	Annual pre-positioning review (no official product) County (sets the date)	PRE-POSITION: confirm shelters/refuge outside the tsunami evacuation zone (and extreme zone), transport plan for no-vehicle residents inside the zone, pre-approved multilingual evacuation messages.	County Emergency Management (each county)	hiema, dhs, dot
DURING	Tsunami Warning NOAA Pacific Tsunami Warning Center (PTWC)	Evacuate the zone; open refuge outside the zone; prioritize residents who cannot evacuate on their own.	County Emergency Management (each county)	hiema, dot, red_cross
AFTER	Warning cancelled or all clear County / NWS / PTWC	Shelter displaced households, feeding, power for medical devices, donations and volunteers.	American Red Cross	hawaii_voad, doh, utilities, salvation_army

Source: National Weather Service, Central Pacific Hurricane Center and Pacific Tsunami Warning Center product definitions (links in section 8).

7. Priority outreach and languages

Top 10 Census areas by number of people with needs

#	Community (Census area)	People in the risk area	Use power-dependent medical equipment (estimate)	People with a disability	Speak English less than well	People in households without a car	Needs total (a person can be in more than one column)
1	Hilo Census area 205	1,344	8	202	36	291	537
2	Kailua Census area 216.08	1,971	16	264	150	27	458
3	Hilo Census area 204	1,284	8	184	30	171	394
4	Hilo Census area 206	1,218	8	226	12	69	315
5	Kailua Census area 215.04	917	4	115	38	66	222
6	Hawaiian Paradise Park Census area 210.14	1,391	8	143	44	0	195
7	Hawaiian Beaches / Nānāwale Estates Census area 211.08	474	3	115	4	37	160
8	Kahalu‘u-Keauhou Census area 215.10	718	7	82	34	28	152
9	Hilo Census area 203	321	2	52	2	76	133
10	Hawaiian Paradise Park Census area 210.15	474	3	119	6	0	128

Source: people and needs – CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER and FEMA National Risk Index v1.20; community names – 2020 Census blocks and places, spellings from USGS GNIS.

Languages: people age 5 and older who speak English less than "very well" (Hawaii)

Language	People	Margin of error (90%)
Ilocano	3,409	±869
Other English-based Creole languages	2,040	±740
Japanese	1,408	±504
Spanish	1,217	±499
Tagalog, Filipino	924	±368
Chinese	651	±326
Hawaiian	637	±264
Marshallese	538	±303

Source: ACS PUMS 2019–2023 5-year (validated against DBEDT Data Book 2024, Table 1.44).

Pre-approved message templates: none yet; no machine translation without human review.

8. Sources and assumptions

Planning factors

Factor	Value	Source or team assumption
Shelter-seeking rate – Hurricane	0.02 / 0.03 / 0.05	South Carolina EMD, 2026 Hurricane Response Annex G (planning factors 2%, 3%, 5%)
Shelter-seeking rate – Wildfire	0.02 / 0.05 / 0.112	high = observed Maui 2023: 1,418 people in shelters on Aug 11, 2023 (Maui Now) / Lahaina CDP population 12,702 (2020 Census) = 0.1116, rounded to 3 decimals = 0.112 (so the calibration event falls inside the band). low and central are team_assumption.
Shelter-seeking rate – Tsunami	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Shelter-seeking rate – Inland flooding	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Shelter-seeking rate – Coastal flooding	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Footprint – Hurricane	all	Hurricanes affect whole islands (Lala, Lowell 2026).
Footprint – Tsunami	tsunami_zone_share	Population share inside the State standard tsunami evacuation zone. Extreme zone is reported separately.
Footprint – Wildfire	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
Footprint – Inland flooding	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
Footprint – Coastal flooding	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
floor sqft 72h	20	Florida DEM Sheltering Guidance Aid (2010): emergency evacuation shelter, <=72 h
floor sqft 2wk	40	Florida DEM (2010): short-term shelter, <=2 weeks
water gal per day	1	Florida DEM (2010)
meals per day	2	Florida DEM (2010): two hot meals a day plus snacks
persons per toilet short term	20	Florida DEM (2010): short-term shelter, <=2 weeks
power outlets per power dependent person	1	team_assumption
persons per toilet 72h	40	Florida DEM (2010): emergency evacuation shelter, <=72 h
Staffing ratios	client registration 1:165, dormitory management 1:100, meal service 1:110, mental health 1:250, public health medical 1:200, security 1:100, sanitation 1:200, children area 1:12	One staff member per N shelter residents, minimum 1 when the shelter is open. Florida DEM (2010) staffing metrics; the source marks them as recommendations.
dog owning household share	0.446	AVMA U.S. pet ownership statistics (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Dog ownership increased by 6% from 38.4% (2016) to 44.6% (2020)'. The first url did not resolve when checked 2026-10-27.
cat owning household share	0.26	AVMA (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Cat ownership increased slightly from 25% in 2016 to 26% in 2020'.

Data sources

Source	Fetched	URL / file
CDC/ATSDR Social Vulnerability Index 2022, census tracts	2026-10-27	https://onemap.cdc.gov/onemapservices/rest/services/SVI/CDC_ATSDR_Social_Vulnerability_Index_2022_USA/FeatureServer/2
FEMA National Risk Index v1.20, census tracts – Hurricane	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Hurricane_Hazard_Type_Risk_Ind

Source	Fetched	URL / file
FEMA National Risk Index v1.20, census tracts – Wildfire	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Wildfire_Hazard_Type_Risk_Index
FEMA National Risk Index v1.20, census tracts – Tsunami	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Tsunami_Hazard_Type_Risk_Inde
FEMA National Risk Index v1.20, census tracts – Inland Flooding	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Inland_Flooding_Hazard_Type_Ris
FEMA National Risk Index v1.20, census tracts – Coastal Flooding	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Coastal_Flooding_Hazard_Type_F
Tsunami Evacuation Zones (Statewide GIS)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/2
Extreme Tsunami Evacuation Zones (Statewide GIS)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/12
HHS emPOWER – electricity-dependent DME, county level	2026-10-27	https://services2.arcgis.com/ZQ4jTQn6k7VPXEwO/arcgis/rest/services/HHS_emPOWER_REST_Service_Public/FeatureServer/2
Hawaii Statewide GIS – State DFIRM (FEMA flood hazard areas, 6/15/26)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/6
County shelter activations 2026 (official county releases)	see table	data/sources/county_shelter_activations.csv
2020 Census Block Assignment File, place/CDP (tract community labels)	2026-10-27	https://www2.census.gov/geo/docs/maps-data/data/baf2020/BlockAssign_ST15_HI.zip
2020 Census PL 94-171 block population P1_001N (tract community labels)	2026-10-27	https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting_File--PL_94-171/Hawaii/hi2020.pl.zip
ACS PUMS 2019-2023 5-year (detailed languages)	2026-10-27	https://www2.census.gov/programs-surveys/acs/data/pums/2023/5-Year/csv_phi.zip
DBEDT State of Hawaii Data Book 2024, Table 1.44 (validation)	2026-10-27	https://files.hawaii.gov/dbedt/economic/databook/2024-individual/01/014424.pdf
Partner list: Hawaii DHS, HAIC 2026 ESF-6 challenge statement	2026-10-15	HAIC_AI Challenge_Submission_DHS_Disaster_2026.pdf
pre season hurricane		https://www.nhc.noaa.gov/cphcmission.php

Source	Fetched	URL / file
hurricane watch		https://www.nhc.noaa.gov/aboutgloss.shtml
hurricane warning		https://www.nhc.noaa.gov/aboutgloss.shtml
tsunami warning		https://www.tsunami.gov/?page=message_definitions
flash flood warning		https://www.weather.gov/safety/flood-watch-warning
red flag warning		https://www.weather.gov/safety/wildfire-ww
coastal flood warning		https://www.weather.gov/okx/marinedef

Privacy: aggregate census-tract and county data only. No individual records. Public facility names and addresses only.